**Pokhara University**
**Faculty of Science and Technology**

Course No.: xxx xxx
Course Title: Applied Physics (3-1-2)
Nature of the Course: Theory and Practical
Year: First
Level: Bachelor

Full Marks: 100
Pass Marks: 45
Time per Period: 1 hour
Total Periods: 45
Program: BE

## 1. Course Description

This course covers the fundamental topics of physics and basic principles that are required to study other engineering courses. It develops the ability to identify, formulate and solve engineering physics problems. Moreover, it enables to formulate, conduct, analyze and interpret experiments in engineering physics through tutorials, laboratory work and self-learning activities.

## 2. General Objectives

The general objectives of this course are:

- To equip the students with the fundamental concept and laws of oscillation, electromagnetism and thermodynamics.
- To acquaint the students with waves, laser, optical fiber, quantum mechanics and enlighten the importance of capacitor and dielectrics.

## 3. Methods of Instruction

Lecture, discussion, tutorials, Laboratory work and assignments

## 4. Contents in Detail

| Specific Objectives | Contents |
|---|---|
| Understand mechanical oscillations, solve problems related to different types of oscillation, familiar with the scope in various engineering fields. | **Unit I: Mechanical Oscillation (6 hrs)**<br>1.1 Free oscillation, Damped oscillation and Forced oscillation (Physical meaning and equations).<br>1.2 Compound pendulum, Minimum and maximum time period in compound pendulum, Interchangeability of point of suspension and point of oscillation in compound pendulum, Torsion pendulum, Determination of modulus of rigidity of material using torsion pendulum. |
| Solve and analyze the problems related to waves | **Unit II: Wave Motion (4 hrs)**<br>Introduction of wave, wave velocity and particle velocity, types of waves and their applications, Speed of wave in stretched string, energy, power and intensity of plane progressive wave, standing wave and resonance, sonometer. |

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| Solve the problems related to reverberation in different units of building. Solve the problems related to ultra sound. | **Unit III: Acoustics (4 hrs.)**<br>3.1 Classification of sound waves, Acoustics of building, Reverberation of sound, absorption coefficient, Noise pollution and its control, Sound insulation, Sabine equation.<br>3.2 Introduction, production and applications of ultrasonic wave. Ultrasonic method in non-destructive testing. |
| understand the use of lasers in engineering sciences and solve problems related to laser and fiber optics. Apply the concept of optical fibers in communication system and sensors. | **Unit IV: Photonics (6 hrs.)**<br>4.1 Laser: Introduction of laser, Principles of generation of laser light (induced absorption, spontaneous emission, stimulated emission, population inversion, pumping, metastable state), He–Ne laser, Semiconductor laser, Applications of laser.<br>4.2 Fiber optics: Introduction, Types of optical fiber, Principle of propagation of light wave through optical fiber (Acceptance angle), Numerical aperture, Applications of optical fiber in communications, Optical fiber sensors. |
| Evaluate the capacity of capacitors to store energy with and without dielectrics. Solve problems related to electrostatics. | **Unit V: Capacitor and Dielectric (6 hrs.)**<br>5.1 Capacitor: Introduction, Types of capacitors, Charging and discharging of capacitor.<br>5.2 Dielectric: Introduction, Dielectric constant, electric flux density, Polarization, Polarization in free space, Gauss law in dielectric, Electronic and Ionic polarization (Clausius-Mossotti equation). |
| Deal with interaction between electric field and magnetic field on matter. Analyze the relationship between electric field, magnetic field and speed of wave. | **Unit VI: Electromagnetism (6 hrs.)**<br>6.1 EM Oscillation: LC oscillation, Damped LCR oscillation, Forced em oscillation, resonance and quality factor<br>6.2 EM waves: Maxwell equations in integral form, Conversion of Maxwell's equations in differential form, Continuity equation, Relation between electric field, magnetic field and speed of light, wave equations in free space, verification of light wave as an electromagnetic wave, Wave equation in dielectric medium |
| Apply principles of quantum mechanics to investigate the observables on known wave functions. Solve the problems | **Unit VII: Quantum Mechanics (5 hrs.)**<br>Inadequacy of classical mechanics, Importance of quantum mechanics, Matter wave (de-Broglie equation), Wave function and its significance, |

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| related to particle wave using Schrodinger's wave equations. | Energy and momentum operator, Time independent and time dependent Schrodinger wave equations, Application of Schrodinger wave equation for the electron in metal, Normalized wave function describing the motion of an electron inside in an infinite potential well. |
| Acquainted with the laws of thermodynamics and applications. Solve the problems related to thermodynamics and heat transfer. | **Unit VIII: Fundamentals of Thermodynamics and Heat Transfer (8 hrs.)**<br>8.1 Concepts and definition: applications of thermodynamics, properties and state of substance, thermodynamics properties and types, processes (definition, characteristics and examples): reversible and irreversible process.<br>8.2 Laws of thermodynamics: first law of thermodynamics, first law for closed system, internal and stored energy, joules law, enthalpy, specific heat, application of first law for closed system, Related problems on closed system, second law of thermodynamics, heat engine (four components of refrigerator and heat pump, COP of refrigerator and heat pumps), Kelvin-Planck and Clausius statement of second law.<br>8.3 Heat transfer: modes of heat transfer (conduction, convection and radiation), statement and assumption of Fourier's law of thermal conductivity, one dimensional steady state heat conduction through plane wall, basic laws of radiation (Emissive power and emissivity, Stefan-Boltzmann's law), Concept of black bodies. |

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**5. Tutorials**

1. Solving the problems related to different oscillation.
2. Solving and analyzing the problems related to waves.
3. Determination of standard reverberation time for normal human ear and solving problems related to ultra sound.
4. Determination of angle of acceptance for working of optical fiber and finding population of atoms in different energy states.
5. Solving the problems for different combination of capacitors and finding the charging and discharging time constant for capacitor.
6. Solving the problems related to Gauss law of electrostatics.
7. Determination of frequency of damped and undammed LC oscillation and analyzing the relationship between electric field, magnetic field and speed of wave.
8. Solving the problems related to particle wave using Schrodinger's wave equations.

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9. Solving the problems related to thermodynamics and heat transfer.

**6. Laboratories (Any Eight):**

1. To determine the acceleration due to gravity and radius of gyration of bar pendulum.
2. To determine the value of modulus of rigidity of the material given and moment of inertia of circular disc using torsion pendulum.
3. To determine the acceptance angle of an optical fiber using laser source.
4. To determine the frequency of AC mains by using sonometer apparatus.
5. To determine the wavelength of laser light by using diffraction grating
6. To determine the capacitance of given capacitor by charging and discharging through resistor.
7. To plot a graph between current and frequency in an LRC series circuit and to find: i) the resonance frequency ii) the quality factor.
8. To determine the dielectric constant of a given material
9. To determine the Planck's constant and photoelectric work functions of the material.
10. To measure the pressure, specific volume and temperature.
11. To find out the efficiency of a compressor.
12. To measure the rate of heat, transfer by conduction
13. To measure the performance of a Refrigeration/ Heat pump

**7. Evaluation system and Students' Responsibilities**

**Evaluation System**

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows.

| **External Evaluation** | **Marks** | **Internal Evaluation** | **Weight** | **Marks** |
|---|---|---|---|---|
| Semester-End examination | 50 | **Theory** | | 30 |
| | | Attendance and Class Participation | 10% | |
| | | Assignments | 20% | |
| | | Presentations/Viva/Quizzes | 10% | |
| | | Term exam | 60% | |
| | | **Practical** | | 20 |
| | | Attendance and Class Participation | 10% | |
| | | Report | 10% | |
| | | Viva | 20% | |
| | | Exam | 60% | |
| | | Total Internal | | 50 |
| Full Marks: 50 + 50 = 100 | | | | |

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**Student Responsibilities**

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given

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NOT QUALIFIED (NQ) and the student will not be eligible to appear the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for re-exam.

## 8. Prescribed Books and References

### Text Books


1. Halliday, D., Resnick, R., & Walker. J. *Fundamental of Physics*. John Wiley and Sons. Inc.
2. Young, H. D. & Freedman. R. A. *Sears and Zemansky's University Physics*. 2009.
3. Howel, J. R. & Buckius, R. O. *Fundamentals of Engineering Thermodynamics*. McGraw-Hill Publishers.


### References


1. Reitz, J., Milford, F.J., Christy, R.W., Foundations of Electromagnetic Theory, 1996.
2. David, J. Griffiths, Introduction to Electrodynamics, Prentice Hall of India Private Limited, New Delhi, 2008
3. Van Wylen, G. J. and Sonntag, R. E., Fundamentals of Classical Thermodynamics, Wiley Eastern Limited, New Delhi, 1989
4. Malik, H. K., Singh, A. K., Engineering Physics, Tata McGraw Hill Education Private Ltd., 2010.
5. Arora, C. L. (2020). *B. Sc. Practical Physics*. S. Chand Publishing.
6. Mathur, D. S., Mechanics, S. Chand and Company Ltd., 2003.
7. Subrahmanyam, N., Lal, B., A text book of Optics, S. Chand and Company Ltd., 2005.
8. Tiwari, K. K., Electricity and Magnetism, S. Chand and Company Ltd., 2001
9. Murugeshan, R., Sivaprasath, K., Modern Physics, S. Chand and Company Ltd., 2009.


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